

Q 6.6.) 만약 $n \times 1$ 벡터 y 가 $N(0, I)$ 의 분포를 한다면 다음의 이차형식은 어떤 분포를 하는가? x 는 $n \times (k+1)$ 행렬이다.

$$(1) y'x(x'x)^{-1}x'y \quad (2) y'[I - x(x'x)^{-1}x']y$$

$$(1) y'x(x'x)^{-1}x'y = y'Ay, \quad A = x(x'x)^{-1}x'$$

$$A \text{ is idempotent} \quad \therefore y'Ay \sim \chi^2(r; \delta), \quad r = \text{rank}(A), \quad \delta = \mu'A\mu = 0 \\ \therefore y'x(x'x)^{-1}x'y \sim \chi^2(k+1)$$

$$(2) y'[I - x(x'x)^{-1}x']y \sim \chi^2(n-k-1)$$

Q 6.7.) 중회귀모형 $y = x\beta + \varepsilon$, $\varepsilon \sim N(0, I)$ 이 성립한다면 다음의 이차형식은 어떤 분포를 하는가? x 는 $n \times (k+1)$ 행렬이다.

$$(1) y'x(x'x)^{-1}x'y = Q_1 \quad (2) y'[I - x(x'x)^{-1}x']y = Q_2$$

(3) Q_1 과 Q_2 가 독립적으로 분포하고 있음을 보려라.

$$(4) \frac{Q_1/(k+1)}{Q_2/(n-k-1)}$$

$$y \sim N(x\beta, I)$$

$$(1) y'x(x'x)^{-1}x'y = Q_1 \sim \chi^2(k+1; \delta), \quad \delta = E[y]A E[y] = \beta'x'x(x'x)^{-1}x'\beta = \beta'x'x\beta \\ \therefore x(x'x)^{-1}x' \cdot \text{Var}(y) = x(x'x)^{-1}x' \cdot I_n \text{ is idempotent.}$$

$$(2) y'[I - x(x'x)^{-1}x']y = Q_2 \sim \chi^2(n-k-1; \delta), \\ \delta = E[y]A E[y] = \beta'x'[I - x(x'x)^{-1}x']x\beta = 0$$

$$\therefore A = [I - x(x'x)^{-1}x'] \text{ is idempotent}$$

$$(3) y'Ay \text{ and } y'By \text{ are independent} \Leftrightarrow A \Sigma B = 0, \quad \Sigma = \text{Var}(y)$$

$$A \Sigma B = [x(x'x)^{-1}x'] \text{Var}(y) [I_n - x(x'x)^{-1}x'] \\ = x(x'x)^{-1}x' - x(x'x)^{-1}x'x(x'x)^{-1}x' \\ = x(x'x)^{-1}x' - x(x'x)^{-1}x' = 0$$

$$\therefore y'(x(x'x)^{-1}x')y \text{ and } y'(I_n - x(x'x)^{-1}x')y \text{ are independent}$$

$$(4) \frac{Q_1/(k+1)}{Q_2/(n-k-1)} \sim F(k+1, n-k-1; \delta) \quad \delta = \beta'x'x\beta$$

Q 6.8.) 만약 $y_j \sim N(\mu_j, \sigma_j^2)$, $j=1, 2, \dots, n$ 이 서로 독립이라고 하자

$$(1) \sum_{j=1}^n \frac{1}{\sigma_j^2} (y_j - y^*)^2 \sim \chi^2(n-1, \lambda) \text{ 임을 증명하라. 여기서 } y^* = \frac{\sum (\frac{y_j}{\sigma_j^2})}{\sum (\frac{1}{\sigma_j^2})}$$

Let $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, $y \sim N(\mu, \Sigma)$, $y \in \mathbb{R}^n$, $\mu = E[y] \in \mathbb{R}^n$, $\Sigma \in \mathbb{R}^{n \times n}$

$$y^* = \frac{\sum (\frac{y_j}{\sigma_j^2})}{\sum (\frac{1}{\sigma_j^2})}, \quad \sum_{j=1}^n \frac{y_j}{\sigma_j^2} = z = [y_1, \dots, y_n] \begin{bmatrix} \sigma_1^{-2} \\ \vdots \\ \sigma_n^{-2} \end{bmatrix} = y' \Sigma^{-1} \mathbf{1} \\ \sum_{j=1}^n \frac{1}{\sigma_j^2} = [1 \dots 1] \begin{bmatrix} \sigma_1^{-2} \\ \vdots \\ \sigma_n^{-2} \end{bmatrix} = [1 \dots 1] \begin{bmatrix} \sigma_1^{-2} & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & \sigma_n^{-2} & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \\ = \mathbf{1}' \Sigma^{-1} \mathbf{1}$$

$$(y - y^*) = (I_n - \Pi_p) y, \quad \Pi_p \text{ is unknown.}$$

$$(y - y^*) = (I_n - \mathbf{1} \frac{\mathbf{1}' \Sigma^{-1}}{\mathbf{1}' \Sigma^{-1} \mathbf{1}}) y$$

$$\sum_{j=1}^n \frac{1}{\sigma_j^2} (y_j - y^*)^2 = y' [I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}] \Sigma^{-1} (I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}) y$$

$$\text{Let } A = [I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}] \Sigma^{-1} [I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}]$$

$$A \Sigma = [I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}] \Sigma^{-1} (\Sigma - \mathbf{1}) \\ = [I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}] (\Sigma - \mathbf{1}) = [I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}] \Sigma$$

$$A \Sigma A \Sigma = [I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}] [I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1}] \Sigma \\ = I_n - 2 \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1} + \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1} \Sigma \\ = I_n - \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1} \mathbf{1}' \Sigma^{-1} = A \Sigma$$

$$\therefore A \Sigma \text{ is idempotent}$$

$$\text{trace}(A) = \text{trace}(I_n) - \text{trace}(\mathbf{1}' \Sigma \mathbf{1} (\mathbf{1}' \Sigma^{-1} \mathbf{1})^{-1}) \\ = n - 1$$

$$\delta = E[y - y^*]' A E[y - y^*] \\ = y' (I - A)' A (I - A) y$$

$$\therefore \sum_{j=1}^n \frac{1}{\sigma_j^2} (y_j - y^*)^2 \sim \chi^2(n-1, \lambda), \quad \lambda = \delta$$

$$(b) \text{ if } \mu_1 = \dots = \mu_n = 0, \quad E[(I_n - A)y] = (I_n - A)E[y] = (I_n - A) \cdot 0 = 0$$

6.9. 독립변수가 둘인 중회귀모형

$$y_j = \beta_0 + \beta_1(x_{1j} - \bar{x}_1) + \beta_2(x_{2j} - \bar{x}_2) + \varepsilon_j, \quad j = 1, 2, \dots, n$$

에서 $\varepsilon \sim N(0, \sigma^2)$ 일 때 β_1 의 최소제곱추정량 b_1 의 분산이 다음과 같음을 보이라.

$$\text{Var}(b_1) = \frac{\sigma^2}{\sum_{i=1}^n (x_{1i} - \bar{x}_1)^2 - r^2}$$

$$y = [y_1 \dots y_n]^T, \quad \beta = (\beta_0 \quad \beta_1 \quad \beta_2)^T, \quad X = (\mathbf{1} \quad x_1 \quad x_2)$$

$$\mathbf{1} = (1, \dots, 1)^T, \quad x_1 = (x_{11}, \dots, x_{1n})^T$$

$$x_2 = (x_{21}, \dots, x_{2n})^T$$

$$\varepsilon = (\varepsilon_1 \dots \varepsilon_n)^T$$

$$\therefore y = \mathbf{1}\beta_0 + (I_n - \Pi_{\mathbf{1}})x_1\beta_1 + (I_n - \Pi_{\mathbf{2}})x_2\beta_2$$

$$y - \hat{y} = y - \Pi_X y$$

$$\therefore L(\beta) = (y - \hat{y})^T (y - \hat{y}) = y^T (I_n - \Pi_X) y = y^T y - y^T \Pi_X y$$

Q.6.10.) 중회귀모형 $y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, I\sigma^2)$ 에서 X 가 $n \times (k+1)$ 행렬이고

계수가 $(k+1)$ 이라면 적합된 모형 $\hat{y} = Xb$ 에 대하여

$$\sum_{j=1}^n \text{Var}(\hat{y}_j) = (k+1)\sigma^2 \quad \text{이 됨을 증명하라}$$

$$\begin{aligned} \sum_{j=1}^n \text{Var}(\hat{y}_j) &= \text{trace}(\text{Var}(\hat{y})) = \text{trace}(\text{Var}(\Pi_X y)) \\ &= \text{trace}(X(X^T X)^{-1} X^T \text{Var}(y) X(X^T X)^{-1} X^T) \\ &= \text{trace}(\sigma^2 X(X^T X)^{-1} X^T) = \text{trace}(\sigma^2 (X^T X)^{-1} X^T X) = \text{tr}(\sigma^2 I_{k+1}) \\ &= \sigma^2 (k+1) \end{aligned}$$

Q.6.11.) 중회귀모형, $y = X\beta + \varepsilon, \quad \varepsilon \sim N(0, I\sigma^2)$ 에서 X 가 $n \times (k+1)$ 행렬이고 계수가

$k+1$ 이라 하자. $X\beta$ 를

$$X\beta = (x_1 \quad x_2) \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$

와 같이 나누어 쓸 때, β_2 의 최소제곱추정량 b_2 는 다음과 같음을 증명하라.

$$b_2 = [x_2' x_2 - x_2' x_1 (x_1' x_1)^{-1} x_1' x_2]^{-1} [x_2' y - x_2' x_1 (x_1' x_1)^{-1} x_1' y]$$

또한 $\text{Var}(b_2)$ 를 구하라.

$$y = X\beta, \quad \hat{y} = X\hat{\beta} = \Pi_X y = \Pi_{x_1} y + \Pi_{x_2, \perp} y, \quad \Pi_{x_2, \perp} = x_{2, \perp} (x_{2, \perp}^T x_{2, \perp})^{-1} x_{2, \perp}^T$$

$$x_{2, \perp} = x_2 - \Pi_{x_1} x_2$$

$$\begin{aligned} \hat{y} &= x_1 \hat{\beta}_1 + x_2 \hat{\beta}_2 = x_1 \hat{\beta}_1 + (\Pi_{x_1} x_2 + x_{2, \perp}) \hat{\beta}_2 \\ &= x_1 \hat{\beta}_1 + \Pi_{x_1} x_2 \hat{\beta}_2 + x_{2, \perp} \hat{\beta}_2 = \Pi_{x_1} y + \Pi_{x_2, \perp} y \end{aligned}$$

$$x_{2, \perp} \hat{\beta}_2 = \Pi_{x_2, \perp} y \Rightarrow \hat{\beta}_2 = (x_{2, \perp}^T x_{2, \perp})^{-1} x_{2, \perp}^T y$$

$$(x_{2, \perp}^T x_{2, \perp})^{-1} = [x_2^T x_2 - x_2^T \Pi_{x_1} x_2]^{-1} \quad \text{and} \quad x_{2, \perp}^T y = [x_2^T y - x_2^T \Pi_{x_1} y]$$

$$\therefore b_2 = [x_2^T x_2 - x_2^T x_1 (x_1' x_1)^{-1} x_1' x_2]^{-1} [x_2^T y - x_2^T x_1 (x_1' x_1)^{-1} x_1' y]$$

$$\begin{aligned} \text{Var}(b_2) &= \text{Var}((x_{2, \perp}^T x_{2, \perp})^{-1} x_{2, \perp}^T y) \\ &= (x_{2, \perp}^T x_{2, \perp})^{-1} x_{2, \perp}^T \text{Var}(y) x_{2, \perp} (x_{2, \perp}^T x_{2, \perp})^{-1} \\ &= (x_{2, \perp}^T x_{2, \perp})^{-1} x_{2, \perp}^T [\sigma^2 I_n] x_{2, \perp} (x_{2, \perp}^T x_{2, \perp})^{-1} \\ &= \sigma^2 (x_{2, \perp}^T x_{2, \perp})^{-1} \end{aligned}$$

Q 6.12.) 양의 정수 (positive integer) 인 k_1 과 k_2 에 대하여

$$E[y_j] = \beta_0 + \beta_1 \cos(2\pi k_1 j/n) + \beta_2 \sin(2\pi k_2 j/n), \quad j = 1, 2, \dots, n$$

이 성립할 때에 β_0, β_1 과 β_2 의 최소제곱추정량을 구하라.

$$y_j = \beta_0 + \beta_1 \cos(2\pi k_1 j/n) + \beta_2 \sin(2\pi k_2 j/n) + \varepsilon_j, \quad \varepsilon_i \sim (0, \cdot)$$

$$y = [y_1, \dots, y_n]^T \quad \text{let } X \in \mathbb{R}^{n \times 3}, \quad X = [\mathbf{1} \quad X_1 \quad X_2]$$

$$\mathbf{1} = (1, \dots, 1)^T, \quad X_1 = [\cos(2\pi k_1 1/n), \dots, \cos(2\pi k_1 n/n)]^T \in \mathbb{R}^n$$

$$X_2 = [\cos(2\pi k_2 1/n), \dots, \cos(2\pi k_2 n/n)]^T \in \mathbb{R}^n$$

$$\beta = [\beta_0 \quad \beta_1 \quad \beta_2]^T \in \mathbb{R}^3, \quad \varepsilon = [\varepsilon_1 \dots \varepsilon_n]^T$$

$$y = X\beta + \varepsilon, \quad \hat{y} = \Pi_X y = \Pi_{\mathbf{1}} y + \Pi_{X_1} y + \Pi_{X_2} y$$

$$\Pi_{\mathbf{1}} = \mathbf{1}(\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}'$$

$$\Pi_{X_1} = X_1(X_1'X_1)^{-1}X_1', \quad X_{1\perp} = X_1 - \Pi_{\mathbf{1}}X_1$$

$$\Pi_{X_2} = X_2(X_2'X_2)^{-1}X_2', \quad X_{2\perp} = X_2 - \Pi_{\mathbf{1}}X_2 - \Pi_{X_1}X_2$$

$$\hat{y} = \mathbf{1}\hat{\beta}_0 + X_1\hat{\beta}_1 + X_2\hat{\beta}_2$$

$$= \mathbf{1}\hat{\beta}_0 + (\Pi_{\mathbf{1}}X_1\hat{\beta}_1 + X_{1\perp}\hat{\beta}_1) + (\Pi_{\mathbf{1}}X_2\hat{\beta}_2 + \Pi_{X_1\perp}X_2\hat{\beta}_2 + X_{2\perp}\hat{\beta}_2)$$

$$= \underbrace{\mathbf{1}\hat{\beta}_0 + \Pi_{\mathbf{1}}X_1\hat{\beta}_1 + \Pi_{\mathbf{1}}X_2\hat{\beta}_2}_{\in C(\mathbf{1})} + \underbrace{(X_{1\perp}\hat{\beta}_1 + \Pi_{X_1\perp}X_2\hat{\beta}_2)}_{\in C(X_{1\perp})} + \underbrace{X_{2\perp}\hat{\beta}_2}_{\in C(X_{2\perp})}$$

$$\therefore \Pi_{X_{2\perp}} y = X_{2\perp}\hat{\beta}_2 \Rightarrow \hat{\beta}_2 = (X_{2\perp}'X_{2\perp})^{-1}X_{2\perp}'y$$

$$\Pi_{X_1\perp} y = (X_{1\perp}\hat{\beta}_1 + \Pi_{X_1\perp}X_2\hat{\beta}_2)$$

$$\therefore \hat{\beta}_1 = (X_{1\perp}'X_{1\perp})^{-1}X_{1\perp}' \{ \Pi_{X_1\perp}y - \Pi_{X_1\perp}X_2\hat{\beta}_2 \}$$

$$= (X_{1\perp}'X_{1\perp})^{-1}X_{1\perp}' \Pi_{X_1\perp} (y - X_2\hat{\beta}_2)$$

$$= (X_{1\perp}'X_{1\perp})^{-1}X_{1\perp}' (y - X_2\hat{\beta}_2)$$

$$\Pi_{\mathbf{1}} y = \mathbf{1}\hat{\beta}_0 + \Pi_{\mathbf{1}}X_1\hat{\beta}_1 + \Pi_{\mathbf{1}}X_2\hat{\beta}_2$$

$$\Rightarrow \mathbf{1}\hat{\beta}_0 = \Pi_{\mathbf{1}} y - \Pi_{\mathbf{1}}X_1\hat{\beta}_1 - \Pi_{\mathbf{1}}X_2\hat{\beta}_2$$

$$\therefore \hat{\beta}_0 = (\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}' \Pi_{\mathbf{1}} (y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2)$$

$$= (\mathbf{1}'\mathbf{1})^{-1}\mathbf{1}' (y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2)$$

$$X = \begin{bmatrix} 1 & \cos(2\pi k_1/n) & \sin(2\pi k_1/n) \\ \vdots & \vdots & \vdots \\ 1 & \cos(2\pi k_1/n) & \sin(2\pi k_1/n) \end{bmatrix}$$

$$\text{그런데 } \sum_{t=1}^n \cos(2\pi k_1 t/n) = \sum_{t=1}^n \sin(2\pi k_2 t/n) = 0 \quad \text{이므로}$$

$$X^T X = \begin{bmatrix} n & 0 & 0 \\ 0 & \sum_{t=1}^n \cos^2(2\pi k_1 t/n) & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 & \sum_{t=1}^n \sin^2(2\pi k_2 t/n) \end{bmatrix}$$

이 된다. 또한,

$$\sum_{t=1}^n \cos^2(2\pi k_1 t/n) = \frac{1}{2} \sum_{t=1}^n [1 + \cos(2\pi(2k_1)t/n)] = \frac{n}{2}$$

$$\sum_{t=1}^n \sin^2(2\pi k_2 t/n) = \frac{1}{2} \sum_{t=1}^n [1 - \cos(2\pi(2k_2)t/n)] = \frac{n}{2}$$

이므로, 다음의 결과를 알 수 있다.

$$b_0 = \bar{y}$$

$$b_1 = \frac{2}{n} \sum_{t=1}^n y_t \cos(2\pi k_1 t/n)$$

$$b_2 = \frac{2}{n} \sum_{t=1}^n y_t \sin(2\pi k_2 t/n)$$

Q6.13) 만약

$$E[y_j] = \beta_0 + \beta_1(x_{1j} - \bar{x}_1) + \beta_2(x_{2j} - \bar{x}_2) \quad j = 1, 2, \dots, n \text{ 인 경우에}$$

β_0, β_1 과 β_2 의 최소제곱추정량이 다음과 같은 이단계절차 (two-stage procedure) 에 의하여 구해질 수 있음을 보여라. (단, $\sum_{j=1}^n (x_{1j} - \bar{x}_1)(x_{2j} - \bar{x}_2) = 0$)

(1) 모형 $E[y_j] = \beta_0 + \beta_1(x_{1j} - \bar{x}_1)$

(2) 위의 (1)에서 얻어진 잔차 (residuals)를 종속변수로 하고 $(x_{2j} - \bar{x}_2)$ 를 독립변수로 한 회귀모형을 적합시켜라.

$$y = 1\beta_0 + (x_1 - \pi_1 x_1)\beta_1 + \varepsilon, \quad \varepsilon \sim (0, \cdot)$$

$$\hat{y} = \pi_x y = (\pi_1 + \pi_{x_{1,\perp}})y = 1\hat{\beta}_0 + (x_1 - \pi_1 x_1)\hat{\beta}_1$$

$$\therefore 1(1'1)^{-1}1'y = 1\hat{\beta}_0$$

$$\pi_{x_{1,\perp}}y = (x_1 - \pi_1 x_1)\hat{\beta}_1$$

$$\Rightarrow \hat{\beta}_0 = \frac{1}{n} 1'y$$

$$\therefore \hat{\beta}_1 = (x_{1,\perp}' x_{1,\perp})^{-1} x_{1,\perp}' y, \quad x_{1,\perp} = x_1 - \pi_1 x_1$$

$$\text{Let } y^* = y - \hat{y}, \quad y^* = (I - \pi_1)x_2\beta_2 = x_{2,\perp}\beta_2$$

$$\therefore \hat{\beta}_2 = (x_{2,\perp}' x_{2,\perp})^{-1} x_{2,\perp}' y^*$$